

Qatar-2b

R-0810

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_230214 · created 2026-07-28T11:32:08+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459989.9514 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.183 +/- 0.016
Transit depth (Rp/Rs)^2	3.36 +/- 0.58 [%]
Orbital Inclination (inc)	89.96 +/- 1.34
Airmass coefficient 1 (a1)	0.994 +/- 0.021
Airmass coefficient 2 (a2)	0.003 +/- 0.024
Scatter in the residuals of the lightcurve fit is	2.09 %
Best Comparison Star	#3 - [618, 184]
Optimal Method	PSF photometry
Transit Duration (day)	0.0779 +/- 0.0014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.183 ± 0.016 vs 0.16327 (deviation 1.23σ)
Duration fit vs published	0.0779 d vs 0.0767 d (deviation 0.88σ)
Mid-time O-C	-1.7 min
Depth SNR	11.4
Residual scatter	2.09 %

Light curve

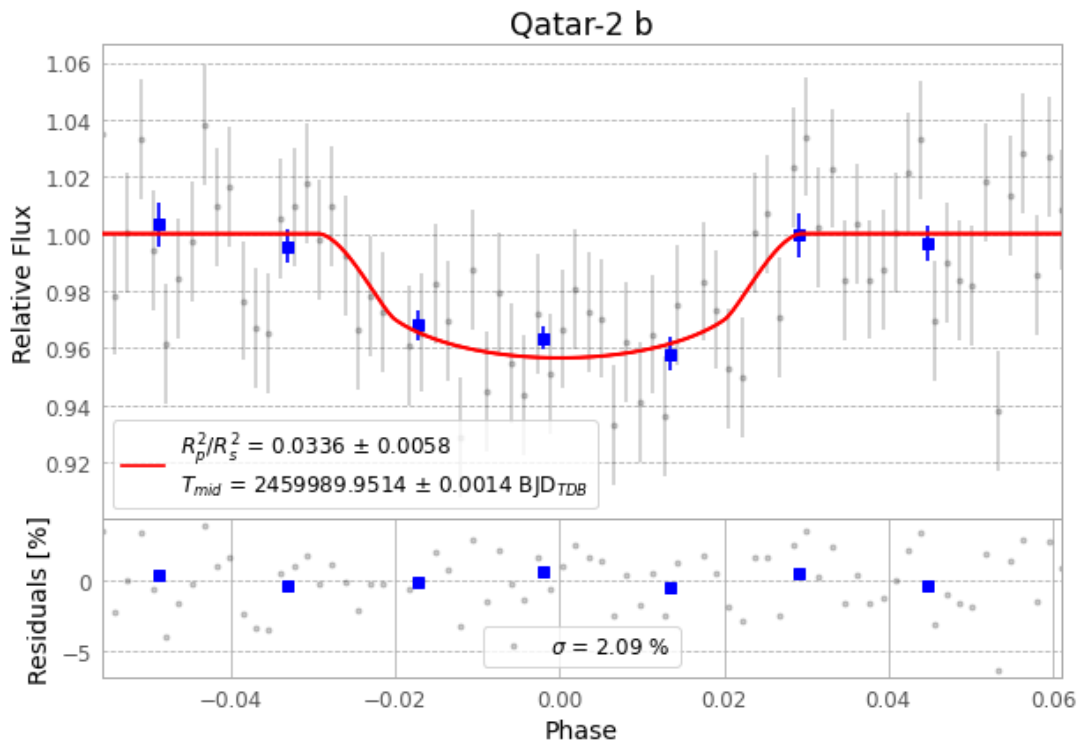


Figure 1 — FinalLightCurve_Qatar-2 b_2023-02-14.png (SHA-256 3bd965273e3308d8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [518, 61], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0cbdb28babaa1170...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

75 FITS frames (49 MB), Qatar-2230214085715.FITS → Qatar-2230214124215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-230214.log (319c74896370...), FinalLightCurve_Qatar-2 b_2023-02-14.png (3bd965273e33...), FinalLightCurve_Qatar-2 b_2023-02-14.csv (e96a488b37d5...)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 11:32Z.